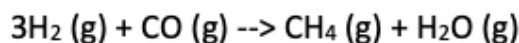


EOX Rate determining step

This is example 5.9 from Rawlings Ekerdt.

The synthesis of methane is carried out over a ruthenium catalyst:



One possible mechanism is

- 1) $\text{CO} (\text{g}) + \text{S} \rightleftharpoons \text{CO}_{\text{ads}}$ θ_{E}
- 2) $\text{CO}_{\text{ads}} + \text{S} \rightleftharpoons \text{C}_{\text{ads}} + \text{O}_{\text{ads}}$ θ_{E}
- 3) $\text{O}_{\text{ads}} + \text{H}_2 (\text{g}) \rightarrow \text{H}_2\text{O} (\text{g}) + \text{S}$ irreversible
- 4) $\text{H}_2 (\text{g}) + 2\text{S} \rightleftharpoons 2\text{H}_{\text{ads}}$ θ_{E}
- 5) $\text{C}_{\text{ads}} + \text{H}_{\text{ads}} \rightleftharpoons \text{CH}_{\text{ads}} + \text{S}$ θ_{E}
- 6) $\text{CH}_{\text{ads}} + \text{H}_{\text{ads}} \rightleftharpoons \text{CH}_{2,\text{ads}} + \text{S}$ θ_{E}
- 7) $\text{CH}_{2,\text{ads}} + \text{H}_{\text{ads}} \rightleftharpoons \text{CH}_{3,\text{ads}} + \text{S}$ θ_{E}
- 8) $\text{CH}_{3,\text{ads}} + \text{H}_{\text{ads}} \rightarrow \text{CH}_4 (\text{g}) + 2\text{S}$ irreversible, RDS

Step 8 is the RDS. Write the rate law for this reaction in terms of only reactant and product species, i.e., do not include any surface concentrations of intermediate species.

$$\text{rate law: } r_{\text{CH}_4} = k_8 \theta_{\text{CH}_3} \theta_{\text{H}}$$

$$\theta_{\text{CH}_3} : \text{use rxn 7}$$

$$0 = k_{7f} \theta_{\text{CH}_2} \theta_{\text{H}} - k_{7r} \theta_{\text{CH}_3} \theta_{\text{S}}$$

$$\theta_{\text{CH}_3} = K_{c7} \frac{\theta_{\text{CH}_2} \theta_{\text{H}}}{\theta_{\text{S}}}$$

$$\theta_{\text{CH}_2} : \text{use rxn 6}$$

$$0 = k_{bf} \theta_{CH} \theta_H - k_{br} \theta_{CH_2} \theta_*$$

$$\theta_{CH_2} = K_{cb} \frac{\theta_{CH} \theta_H}{\theta_*}$$

$\theta_{CH} \circ$ use rxn 5 $^\circ$

$$0 = k_{sf} \theta_c \theta_H - k_{sr} \theta_{CH} \theta_*$$

$$\theta_{CH} = K_{cs} \frac{\theta_c \theta_H}{\theta_*}$$

$\theta_c \circ$ use rxn 2 $^\circ$

$$0 = k_{2f} \theta_{co} \theta_* - k_{2r} \theta_c \theta_o$$

$$\theta_c = K_{c2} \frac{\theta_{co} \theta_*}{\theta_o}$$

$\theta_{co} \circ$ use rxn 1 $^\circ$

$$0 = k_{1f} C_{co} \theta_* - k_{1r} \theta_{co}$$

$$\theta_{co} = K_{c1} C_{co} \theta_*$$

θ_o : Assume $r_3 = r_8$ which is reasonable since $H_2O(g) \frac{1}{2} CH_4(g)$ are produced in stoichiometric amounts.

$$k_{3f} \theta_o C_{H_2} = k_{8f} \theta_{CH_3} \theta_H$$

$$\theta_o = \frac{k_{8f}}{k_{3f}} \frac{\theta_{CH_3} \theta_H}{C_{H_2}}$$

restate rate law : $r_{CH_4} = k_{8f} \theta_{CH_3} \theta_H$

develop expression for θ_{CH_3} :

$$\theta_{CH_3} = k_{c7} \frac{\theta_{CH_2} \theta_H}{\theta_{\#}}$$

sub in θ_{CH_2}

$$\Theta_{CH_3} = K_{c7} K_{c6} \frac{\Theta_{CH} \Theta_H^2}{\Theta_X^2}$$

sub in Θ_{CH} :

$$\Theta_{CH_3} = K_{c7} K_{c6} K_{c5} \frac{\Theta_C \Theta_H^3}{\Theta_X^3}$$

sub in Θ_C :

$$\Theta_{CH_3} = K_{c7} K_{c6} K_{c5} K_{c2} \frac{\Theta_{CO} \Theta_H^3}{\Theta_O \Theta_X^2}$$

sub in Θ_{CO} :

$$\Theta_{CH_3} = K_{c7} K_{c6} K_{c5} K_{c2} \frac{K_{2f}}{K_{2r}} \frac{C_{H_2} \Theta_{CO} \Theta_H^2}{\Theta_{CH_3} \Theta_X^2}$$

re arrange:

$$\Theta_{CH_3}^2 = K_{c7} K_{c6} K_{c5} K_{c2} \frac{K_{2f}}{K_{2r}} \frac{C_{H_2} \Theta_{CO} \Theta_H^2}{\Theta_X^2}$$

sub in $\Theta_{CO} \frac{1}{2} \Theta_H \frac{1}{2}$

θ_H : get from rxn 4°

$$0 = k_{4f} C_{H_2} \theta_H^2 - k_{4r} \theta_H^2$$

$$\theta_H = \theta_H^* \sqrt{k_{4f} C_{H_2}}$$

going back to θ_{CH_3} °

$$\theta_{CH_3}^2 = \frac{k_{3f}}{k_{8f}} k_{c7} k_{c6} k_{c5} k_{c2} C_{H_2}^2 k_{c1} C_{co} k_{c4} \theta_H^*$$

combine°

$$k' = \left(\frac{k_{3f}}{k_{8f}} k_{c7} k_{c6} k_{c5} k_{c2} k_{c1} k_{c4} \right)^{1/2}$$

simplify°

$$\theta_{CH_3} = k' C_{H_2} C_{co}^{1/2} \theta_H^{1/2}$$

substitute°

$$r_{CH_4} = k_{8f} \theta_{CH_3} \theta_H$$

$$= k_{8f} k' C_{H_2} \theta_H C_{co}^{1/2} \theta_H^{1/2}$$

Sub in θ_H

$$r_{CH_4} = K_{eff} K' K_{CH_4}^{1/2} C_{H_2}^{3/2} C_{CO}^{1/2} \theta_*$$

Combine:

$$K'' = K_{eff} K' K_{CH_4}^{1/2}$$

$$r_{CH_4} = K'' C_{H_2}^{3/2} C_{CO}^{1/2} \theta_*$$

θ_* : get from site balance

$$\theta_{tot} = \theta_* + \theta_{CO} + \cancel{\theta_O} + \cancel{\theta_H} + \cancel{\theta_{CH_4}} + \cancel{\theta_{CH_2}} \\ + \cancel{\theta_{CH_3}} + \cancel{\theta_C} \quad \text{assumptions}$$

$$\text{then } \theta_{tot} = \theta_* + K_{C1} P_{CO} \theta_*$$

$$\theta_* = \frac{\theta_{tot}}{1 + K_{C1} C_{CO}}$$

assume $\theta_{co} \gg \theta^*$. Then $K_{c1} C_{co} \gg 1$, so then:

$$\theta^* = \frac{\theta_{tot}}{K_{c1} C_{co}}$$

Substitute: $r_{CH_4} = k'' C_{H_2}^{3/2} C_{co}^{1/2} \theta^{3/2}$

$$r_{CH_4} = \frac{k'' \theta_{tot}^{3/2} C_{H_2}^{3/2}}{K_{c1}^{3/2} C_{co}} = \frac{k_{eff} C_{H_2}^{3/2}}{C_{co}}$$